

6.2: lin. op. $f: V \rightarrow V$; $V = \text{Span}\{(2, 1, 0, 1)^T, (1, 0, 1, 2)^T\} \subseteq \mathbb{R}^4$
splňuje:

$$f\left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 4 \\ 1 \\ 5 \\ 1 \end{pmatrix}, \quad f\left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}$$

Spočítejte $f^n\left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}\right)$

Vezmu $B = ((2, 1, 0, 1)^T, (1, 0, 1, 2)^T) = (b_1, b_2)$.

Poté $f(b_1) = b_1 + 2b_2$, $f(b_2) = 2b_1 - 2b_2$,

tedy $[f]_B^B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$. Najdeme vl. čísla a vektory

této matice:

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = (1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6 \stackrel{\text{chei}}{=} 0$$

$$(\lambda+3)(\lambda-2) = 0 \Rightarrow \lambda_1 = -3 \quad \lambda_2 = 2$$

pro λ_1 :

$$\left(\begin{array}{cc|c} 4 & 2 & 0 \\ 2 & 1 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right) \Rightarrow M_{\lambda_1} = \left\{ p \cdot \begin{pmatrix} 1 \\ -2 \end{pmatrix} \mid p \in \mathbb{R} \right\}$$

pro λ_2 :

$$\left(\begin{array}{cc|c} -1 & 2 & 0 \\ 2 & -4 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right) \Rightarrow M_{\lambda_2} = \left\{ p \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} \mid p \in \mathbb{R} \right\}$$

Když vytvoříme bázi $\bar{B} = \left(1 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} - 2 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}, 2 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right) =$
 $= \left(\begin{pmatrix} 0 \\ -2 \\ -3 \\ 1 \end{pmatrix}, \begin{pmatrix} 5 \\ 1 \\ 1 \\ 4 \end{pmatrix} \right)$, pak $[f]_{\bar{B}}^{\bar{B}} = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$, takže $[f^n]_{\bar{B}}^{\bar{B}} = \begin{pmatrix} (-3)^n & 0 \\ 0 & (2)^n \end{pmatrix}$

Dále $\left[\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right]_{\bar{B}} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Dostáváme

$$[f^n\left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}\right)]_{\bar{B}} = \begin{pmatrix} -1 \cdot (-3)^n \\ (2)^n \end{pmatrix}.$$

Abych dostal souřadnice v $\mathbb{R}^4$, přenásobím získaný vektor vektory báze $\bar{B}$.

$$f^n\left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}\right) = (-3)^n \begin{pmatrix} 0 \\ -2 \\ -3 \\ 1 \end{pmatrix} + (2)^n \begin{pmatrix} 5 \\ 1 \\ 1 \\ 4 \end{pmatrix}$$